SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4314

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage. ●

A registration form must securely send user data to the server.

- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP
C. SMTP
D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer. 2. Identify the **appropriate HTTP method for form submission** and explain why. 3. Choose the **proper semantic element for navigation** and justify its usage. 4. Identify the **correct XML syntax** and explain the error in other options. 5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I Challa priyadarshini(192311118)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Challa Priyadarshini

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	2.5	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	2.3	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	1.8	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	1.8	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	1.0	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
Total	9.4/10				

1. Select the correct HTML tag for video embedding and justify your answer

Correct answer: Option B — <video src="video.mp4" controls></video>

Explanation

The <video> element is the standard HTML5 element used to embed video content directly into a web page. It enables a browser to present video content as part of the page rather than requiring the user to open a separate media player. For the startup, this element is suitable for displaying a promotional video on the homepage so visitors can learn about the company, its products, or its services.
In the given code, the src attribute points to the video file named video.mp4. The controls attribute instructs the browser to display the default video controls. Depending on the browser, these controls allow users to play or pause the video, change the volume, move to a different position, and view playback progress. The opening and closing video tags define the element that contains the video player.

Example code

```
<video controls width="640" poster="preview.jpg">
   <source src="video.mp4" type="video/mp4">
   Your browser does not support HTML video.
</video>
```

This example adds a width attribute to set the player width and a poster attribute to display a preview image before the video begins. The <source> element specifies the video file and its media type. The text inside the video element acts as fallback content for browsers that cannot display HTML video.

Why Option B is correct

- It is a standard HTML5 element supported by modern browsers.
- It can display video content directly within a web page.
- The src attribute identifies the media file.
- The controls attribute provides user-friendly playback controls.
- It supports additional attributes such as width, height, poster, autoplay, muted, loop, and preload, where appropriate.

Why the other options are incorrect

Option A — <media src="video.mp4"></media>: The element <media> is not a standard HTML element for embedding a video. Browsers will not treat it as the built-in HTML video player.
Option C — <movie src="video.mp4"></movie>: The element <movie> is not a standard HTML element. It does not provide the standard video playback behavior.
Option D — <vid src="video.mp4"></vid>: The element <vid> is not a standard HTML video element. The correct element name is <video>.

Best practices

Developers should choose video formats supported by the target browsers and may provide multiple <source> elements to improve compatibility. Videos should be compressed to reduce loading time, given

meaningful accessible controls, and designed to fit mobile screens. Autoplay should be used carefully because it can surprise users and may be blocked by browsers. Captions or subtitles should be provided when speech is important so users who are deaf or hard of hearing can understand the content.
Therefore, Option B is correct because <video> is the proper HTML5 element for embedding a promotional video and the controls attribute allows visitors to control playback.

2. Identify the appropriate HTTP method for form submission and explain why

Correct answer: Option B — POST.

Explanation

HTTP stands for Hypertext Transfer Protocol. It defines how a client, such as a browser, sends requests to a web server and how the server returns responses. HTTP methods indicate the intended action of a request. In the given scenario, the startup has a registration form that sends user information to the server. POST is generally the most appropriate choice for this type of submission because the submitted form data is carried in the request body rather than normally being appended to the URL.
Registration forms may contain a user name, email address, phone number, and password. If GET were used, submitted values can appear in the URL as query parameters and may be stored in browser history, server logs, or copied links. POST avoids placing ordinary form fields in the URL. However, POST alone does not encrypt the information; HTTPS and appropriate server-side security are still required.

Example HTML form

```
<form action="/register" method="post">
   <label for="username">User name:</label>
   <input type="text" id="username" name="username" required>

   <label for="email">Email address:</label>
   <input type="email" id="email" name="email" required>

   <button type="submit">Register</button>
</form>
```

In this example, action="/register" specifies the server endpoint that receives the form. method="post" tells the browser to submit the form using POST. The name attributes identify the values sent to the server, while required asks the browser to check that those fields are filled in before submission. Client-side validation improves the user experience, but the server must validate all received data independently.

Comparison of the HTTP methods

Method	Common purpose	Suitability for registration
GET	Retrieve a resource or information.	Usually unsuitable for submitting private registration data because parameters may be placed in the URL.

POST	Submit data for processing or create a resource through a server endpoint.	Best option among the choices for a standard registration form.
PUT	Create or replace the state of a resource at a known URI, depending on the API design.	Not the usual method for a conventional HTML registration form.
DELETE	Request removal of a resource.	Not appropriate for creating a registration.

Why POST is preferred

- The form values are normally sent in the request body instead of being appended to the URL.
- It is appropriate for requests that submit information for server-side processing.
- It can carry form fields and larger request bodies, subject to server and application limits.
- It is commonly used for account registration, sign-in workflows, contact forms, and creating records.
- It allows the server to validate the input, check business rules, and save permitted data.

Security measures for registration

- Use HTTPS so data is encrypted in transit between the browser and the server.
- Validate and normalize inputs on the server; do not rely only on HTML validation.
- Store passwords using a suitable password-hashing algorithm with a unique salt; never store passwords as plain text.
- Use appropriate CSRF protection where the authentication and session design requires it.
- Limit repeated requests and apply abuse prevention where necessary.
- Return clear error messages without exposing internal server details or sensitive information.

In conclusion, Option B (POST) is the best answer because it is designed for submitting data to be processed by the server. To make the operation secure, the application must additionally use HTTPS and follow secure development practices.

3. Choose the proper semantic element for navigation and justify its usage

Correct answer: Option B — <nav>

Explanation

Semantic HTML uses elements whose names describe the meaning or purpose of their content. The <nav> element represents a section of a page that contains major navigation links. It is suitable for a website menu that helps visitors move between important areas such as Home, About Us, Services, Products, and Contact. Using <nav> makes the structure more meaningful than grouping the same links inside a generic .

Browsers can render a <nav> element like other HTML containers, but assistive technologies can use its semantic meaning to help users identify navigation landmarks. Search engines and other tools can also better interpret the structure of the document. Semantic HTML does not automatically make a site accessible, but it provides a better foundation for accessible navigation.

Example HTML

```
<nav aria-label="Main navigation">
  <ul>
    <li><a href="/">Home</a></li>
    <li><a href="/about">About Us</a></li>
    <li><a href="/services">Services</a></li>
    <li><a href="/contact">Contact</a></li>
  </ul>
</nav>
```

The <nav> element groups the main navigation links. The unordered list (<ul>) and list items (<li>) provide a clear structure, while each anchor (<a>) points to a destination. The aria-label attribute gives the navigation landmark a descriptive name, which is particularly useful when a page contains more than one navigation region.

Why the other options are less suitable

Option A — : This is a generic block-level container. It has no built-in meaning that says its contents are navigation links. It is useful for grouping elements when no more specific semantic element is appropriate.
Option C — <section>: This represents a thematic grouping of related content, usually with a heading. It is not specifically intended to mark a navigation menu.
Option D — <header>: This represents introductory content for a page or section and may contain a logo, heading, or navigation. However, <nav> is the more specific element for identifying a major navigation area.

Benefits of semantic navigation

- Improves the logical structure and readability of HTML.
- Helps screen-reader users identify navigation landmarks.
- Makes the purpose of the navigation area clearer to developers maintaining the website.
- Encourages consistent markup across pages.
- Works with ordinary CSS and JavaScript while preserving the meaning of the content.

Developers should use <nav> for major navigation blocks, not necessarily for every individual link on the page. Navigation should also be keyboard-accessible, have visible focus indicators, use meaningful link text, and adapt to mobile screen sizes. The correct answer is Option B because <nav> directly identifies the purpose of the navigation menu.

4. Identify the correct XML syntax and explain the error in other options

Correct answer: Option B — <note><to>User</to><from>Admin</from></note>

Explanation

XML stands for Extensible Markup Language. It is used to represent structured data in a text format and is often used for configuration files, data exchange, and document formats. Unlike HTML, XML requires

elements to be properly nested and closed. Element names are case-sensitive, and every document must have exactly one root element. A document that follows these structural rules is described as well-formed XML.
Option B is correct because the <note> element is the single root element. The <to> and <from> elements are nested inside it, and each opening tag has a matching closing tag. The child elements are closed in the reverse order in which they were opened, so the document has valid nesting.

Correct XML example

```
<?xml version="1.0" encoding="UTF-8"?>
<note>
   <to>User</to>
   <from>Admin</from>
</note>
```

The XML declaration at the beginning is optional in many XML contexts, but it identifies the XML version and character encoding. The root element <note> contains two child elements: <to> and <from>. Both child elements are closed correctly, and then the root element is closed.

Errors in the other options

Option A — <note><to>User</to><from>Admin</from>: The <note> root element is opened but never closed. The missing </note> end tag makes the document not well-formed.
Option C — <note><to>User</from></note>: The opening tag is <to>, but the closing tag is </from>. XML requires the opening and closing tag names to match exactly, so this is an error.
Option D — <note to="User"></note: The closing tag is incomplete because it is missing the final > character. The correct closing tag would be </note>.

Important XML rules

- Every XML document must contain exactly one root element that encloses the other elements.
- Every non-empty element must have a matching closing tag, or it must be written as a valid self-closing element such as <item />.
- Elements must be properly nested; tags cannot overlap.
- XML names are case-sensitive, so <Name> and <name> are different names.
- Attribute values must be enclosed in quotation marks, such as id="10".
- Reserved characters such as < and & must be escaped when they are intended as text; for example, use < and &.
- A well-formed XML document may still fail validation if it does not conform to an associated schema or document type definition.

Why correct XML matters

Configuration files often store application settings, service endpoints, or other structured information. An XML syntax error may prevent a parser from reading the file, which can stop a program from starting or cause configuration loading to fail. Developers should use an XML-aware editor or parser to validate files

and should distinguish between well-formedness and validation: well-formedness checks basic XML structure, while validation checks whether the content follows a defined schema.
Therefore, Option B is correct because it contains a single root element, properly nested child elements, and matching opening and closing tags.

5. Choose the correct protocol for web communication and justify your choice

Correct answer: Option B — HTTP.

Explanation

HTTP stands for Hypertext Transfer Protocol. It is an application-layer protocol used for communication between web clients and web servers. When a user enters a website address, clicks a link, loads an image, or submits a form, the browser can send an HTTP request to a server. The server processes the request and returns an HTTP response containing a status code and, when appropriate, content such as HTML, JSON, images, or other resources.
HTTP follows a request-response model. A request typically includes a method, a target URL, headers, and sometimes a body. The response contains a status code, headers, and sometimes a body. For example, a successful page request may return a 200 status code, while a missing resource may return 404. HTTP supports methods such as GET and POST, which are used for different purposes in a web application.

Why HTTP is correct

- It is the standard application-layer protocol used to transfer web resources.
- It supports communication between browsers, web applications, and servers.
- It works with methods, headers, status codes, and message bodies to structure requests and responses.
- It can carry HTML pages, images, scripts, stylesheets, JSON data, and other content.
- HTTPS adds transport encryption and server authentication through TLS, making it appropriate for sensitive web traffic.

Why the other protocols are incorrect

Option A — FTP (File Transfer Protocol): FTP is designed for transferring files between systems. It is not the general protocol used for loading ordinary web pages.
Option C — SMTP (Simple Mail Transfer Protocol): SMTP is primarily used to send email between mail clients and mail servers or between mail servers.
Option D — SNMP (Simple Network Management Protocol): SNMP is used to monitor and manage network devices such as routers, switches, and servers.

HTTP and HTTPS

HTTP is the expected answer when the question asks for the standard protocol used for web communication. However, ordinary HTTP does not by itself encrypt traffic. HTTPS is HTTP carried over a connection protected by TLS. TLS helps protect data against eavesdropping and tampering and allows the browser to authenticate the server using its certificate. A modern registration page should use HTTPS, especially because it handles personal information and may transmit credentials.

Using HTTPS is only one part of application security. The application should also validate user input, use secure session management, apply access controls, keep software updated, and avoid exposing sensitive data in URLs or logs. Developers should configure HTTPS correctly and redirect HTTP requests to HTTPS where appropriate.
Therefore, Option B is correct because HTTP is the standard protocol for web requests and responses. In a real production application, HTTPS should be used to protect communication.

Summary of Correct Answers

Question	Correct option	Technology	Main reason
1. Video embedding	B	<video>	HTML5 element that embeds video and can display playback controls.
2. Registration form	B	POST	Submits form data in the request body; HTTPS is needed for encryption.
3. Navigation menu	B	<nav>	Semantic element that identifies a major navigation section.
4. XML syntax	B	Well-formed XML	One root element, correctly nested children, matching tags.
5. Web protocol	B	HTTP	Standard request-response protocol for web communication; HTTPS adds TLS protection.

Overall Conclusion

The scenario demonstrates why web development standards are important when building a reliable modern application. The HTML5 <video> element provides a standard way to display promotional media. The POST method is suitable for sending registration form data to a server, although HTTPS is required to protect the information in transit. The <nav> element gives navigation menus clear semantic meaning and supports more accessible page structure. Correctly formatted XML ensures that configuration data can be parsed, while HTTP provides the request-response foundation for communication between web clients and servers.
The correct options are B for all five questions. Nevertheless, each answer represents a different rule: Option B in the video question is the HTML5 <video> element; Option B in the form question is POST; Option B in the navigation question is <nav>; Option B in the XML question is the well-formed XML snippet; and Option B in the protocol question is HTTP. By using these standards correctly and following additional security and accessibility practices, the startup can build a web application that is easier to maintain, more compatible with browsers, and safer for its users.